Generative Ai And Ai Platform Engineering

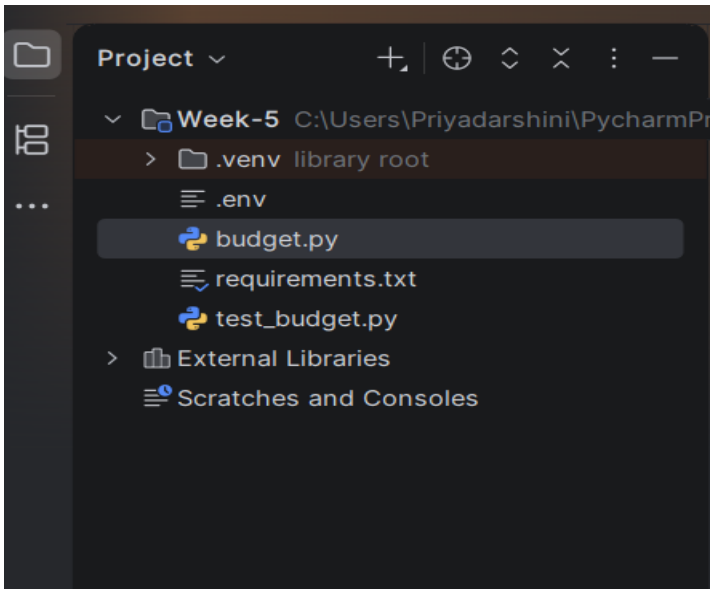
Name:- K. Priyadarshini

Roll no:- 2520030067

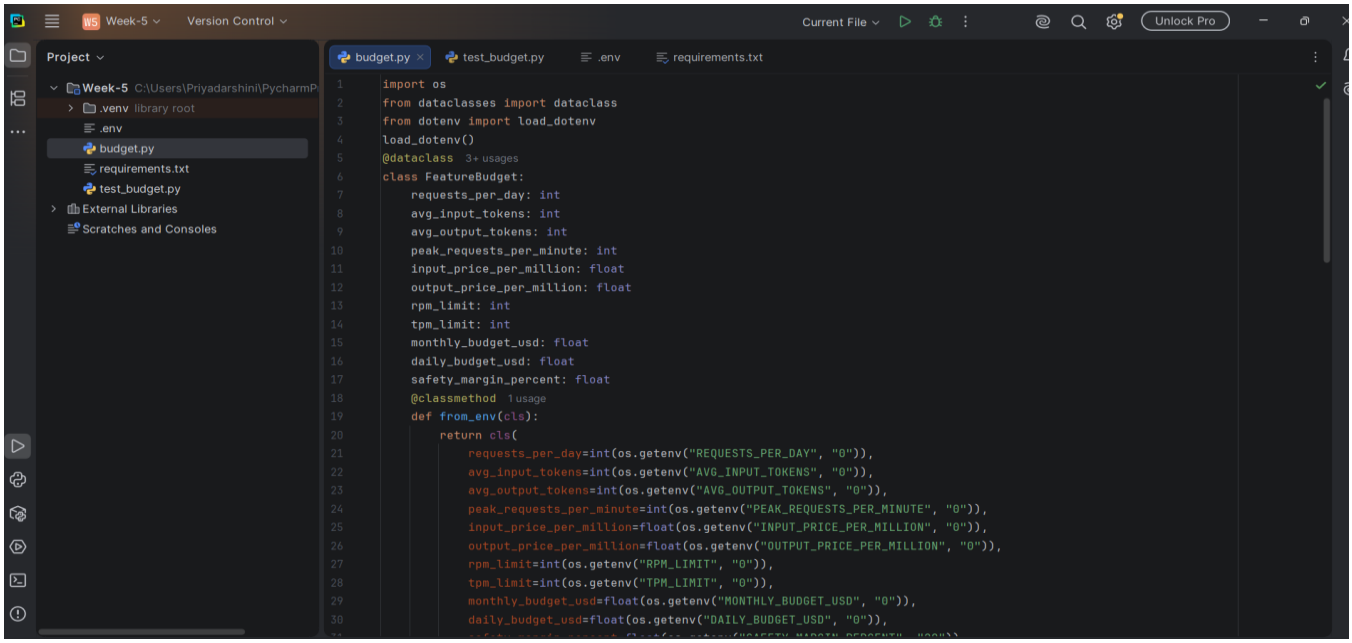
Section no:- 3

WEEK-5

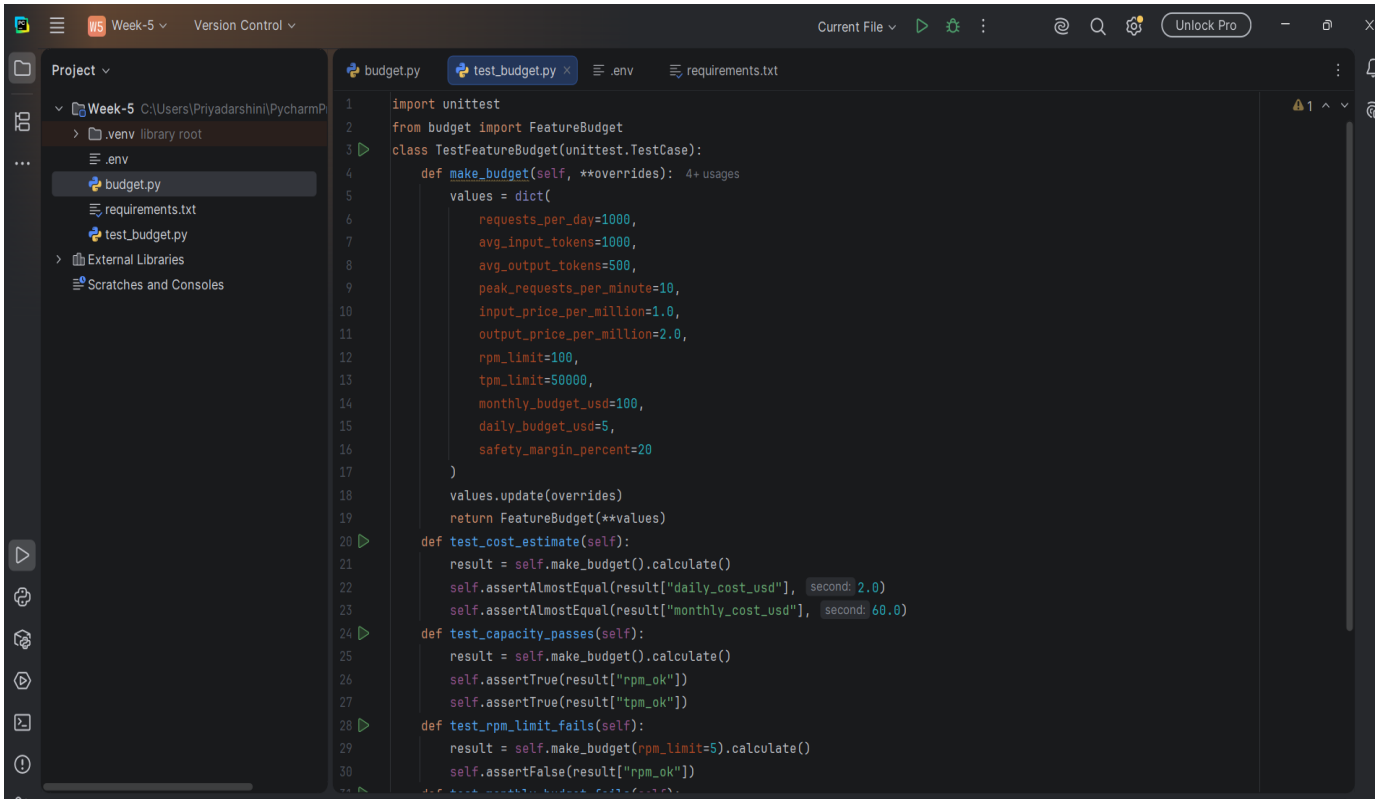
Step 1:-



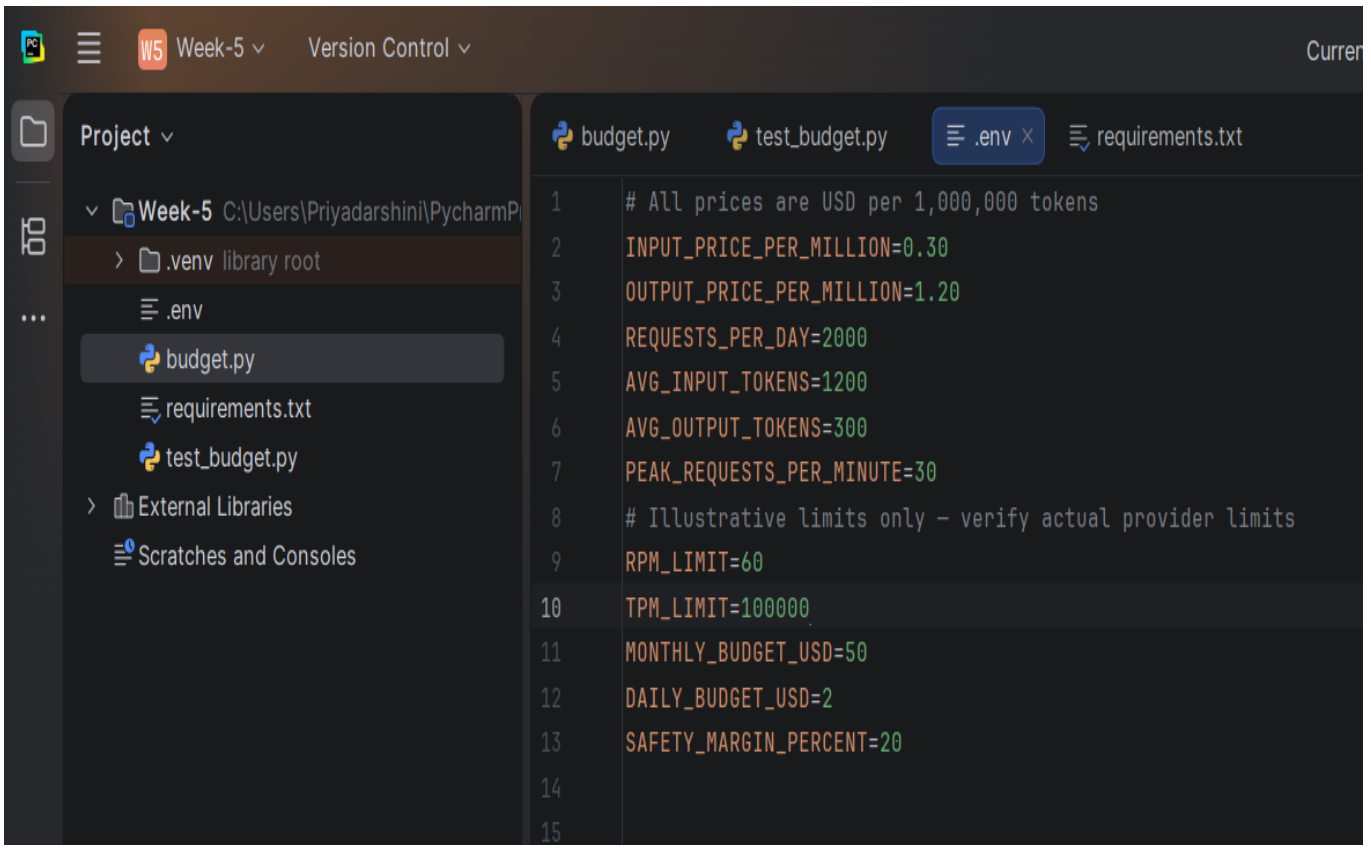
Step 2:-



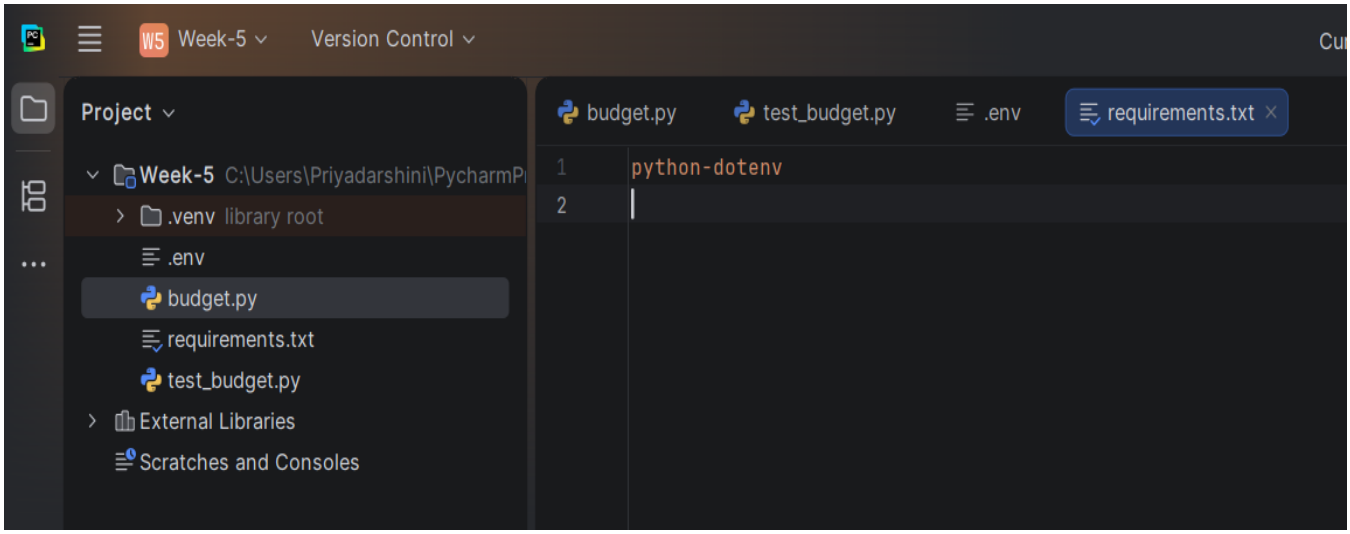
Step 3:-


Week-5
Version Control
Current File
Unlock Pro
Project
budget.py
test_budget.py
.env
requirements.txt
Week-5 C:\Users\Priyadarshini\PycharmP
.venv library root
.env
budget.py
requirements.txt
test_budget.py
External Libraries
Scratches and Consoles
import unittest
from budget import FeatureBudget
class TestFeatureBudget(unittest.TestCase):
def make_budget(self, **overrides): 4+ usages
values = dict(
requests_per_day=1000,
avg_input_tokens=1000,
avg_output_tokens=500,
peak_requests_per_minute=10,
input_price_per_million=1.0,
output_price_per_million=2.0,
rpm_limit=100,
tpm_limit=50000,
monthly_budget_usd=100,
daily_budget_usd=5,
safety_margin_percent=20
)
values.update(overrides)
return FeatureBudget(**values)
def test_cost_estimate(self):
result = self.make_budget().calculate()
self.assertAlmostEqual(result["daily_cost_usd"], second: 2.0)
self.assertAlmostEqual(result["monthly_cost_usd"], second: 60.0)
def test_capacity_passes(self):
result = self.make_budget().calculate()
self.assertTrue(result["rpm_ok"])
self.assertTrue(result["tpm_ok"])
def test_rpm_limit_fails(self):
result = self.make_budget(rpm_limit=5).calculate()
self.assertFalse(result["rpm_ok"])

Step 4:-


Week-5
Version Control
Project
budget.py
test_budget.py
.env
requirements.txt
Week-5 C:\Users\Priyadarshini\PycharmP
.venv library root
.env
budget.py
requirements.txt
test_budget.py
External Libraries
Scratches and Consoles
All prices are USD per 1,000,000 tokens
INPUT_PRICE_PER_MILLION=0.30
OUTPUT_PRICE_PER_MILLION=1.20
REQUESTS_PER_DAY=2000
AVG_INPUT_TOKENS=1200
AVG_OUTPUT_TOKENS=300
PEAK_REQUESTS_PER_MINUTE=30
Illustrative limits only - verify actual provider limits
RPM_LIMIT=60
TPM_LIMIT=100000
MONTHLY_BUDGET_USD=50
DAILY_BUDGET_USD=2
SAFETY_MARGIN_PERCENT=20

Step 5:-



OUTPUT:-

